

```
server {
    listen      1380;
    listen      [::]:1380;
    server_name _;
    root        /opt/myapp/public;

    # Load configuration files for the default server block.
    include /etc/nginx/default.d/*.conf;
}
```

Пытаемся сделать так, чтобы nginx слушала 1380 порт и отдавала страницы из директории /opt/myapp/public

```
[redos@Redos ~]$ sudo systemctl start nginx
[sudo] password for redos:
Sorry, try again.
[sudo] password for redos:
Sorry, try again.
[sudo] password for redos:
[redos@Redos ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/service.d
            └─10-timeout-abort.conf
   Active: active (running) since Tue 2026-04-21 17:34:11 MSK; 8min ago
     Process: 7826 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 7827 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 7828 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 7829 (nginx)
      Tasks: 3 (limit: 2288)
     Memory: 3.7M
        CPU: 24ms
    CGroup: /system.slice/nginx.service
            └─7829 "nginx: master process /usr/sbin/nginx"
              └─7830 "nginx: worker process"
                └─7831 "nginx: worker process"

Apr 21 17:34:11 Redos systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Apr 21 17:34:11 Redos nginx[7827]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Apr 21 17:34:11 Redos nginx[7827]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Apr 21 17:34:11 Redos systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
```

nginx запустился, но не заработал, как нам надо

```
[redos@Redos ~]$ sudo ss -tlnp | grep nginx
LISTEN 0      511          0.0.0.0:80      0.0.0.0:*      users:((("nginx",pid=7831,fd=8),("nginx",pid=7830,fd=8),("nginx",pid=7829,fd=8)))
LISTEN 0      511          [::]:80        [::]:*         users:((("nginx",pid=7831,fd=9),("nginx",pid=7830,fd=9),("nginx",pid=7829,fd=9)))
```

Поняли, что nginx не слушает 1380 порт

```
see systemctl status nginx.service and journalctl -xe nginx.service for details.
[redos@Redos ~]$ sudo semanage port -a -t http_port_t -p tcp 1380
[redos@Redos ~]$ sudo semanage port -l | grep http_port_t
http_port_t                 tcp      1380, 80, 81, 443, 488, 8008, 8009, 8443, 9000
pegasus_http_port_t         tcp      5988
[redos@Redos ~]$
```

Разрешаем веб серверу слушать порт 1380

```
[redos@Redos ~]$ curl http://localhost:1380/
<h1>hello world</h1>
```

Убедились, что теперь nginx корректно настроен

```
[redos@Redos ~]$ sudo systemctl restart nginx
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
```

изменил пути для файлов, куда записываются ошибки и куда записываются журнал доступа, но nginx не запускается

```
[redos@Redos ~]$ sudo chown nginx:nginx /var/log/myapp
[redos@Redos ~]$ sudo systemctl start nginx
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
[redos@Redos ~]$ sudo semanage fcontext -a -t httpd_log_t "/var/log/myapp(/.*)?"
[redos@Redos ~]$ sudo restorecon -Rv /var/log/myapp/
Relabeled /var/log/myapp from unconfined_u:object_r:var_log_t:s0 to unconfined_u:object_r:httpd_log_t:s0
Relabeled /var/log/myapp/access.log from unconfined_u:object_r:var_log_t:s0 to unconfined_u:object_r:httpd_log_t:s0
Relabeled /var/log/myapp/error.log from unconfined_u:object_r:var_log_t:s0 to unconfined_u:object_r:httpd_log_t:s0
[redos@Redos ~]$ ls -ldZ /var/log/myapp
drwxr-xr-x. 2 nginx nginx unconfined_u:object_r:httpd_log_t:s0 4096 Apr 24 11:54 /var/log/myapp
[redos@Redos ~]$ sudo systemctl start nginx
[redos@Redos ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/service.d
           └─10-timeout-abort.conf
   Active: active (running) since Fri 2026-04-24 12:05:43 MSK; 22s ago
```

Поменяли владельца директории, добавили правило в политику SELinux, согласно которому все файлы в директории /var/log/myapp должны получать тип httpd_log_t, после чего nginx запустился

```
[redos@Redos ~]$ ls /var/log/myapp
access.log error.log
[redos@Redos ~]$ curl http://localhost:1380/
<h1>hello world</h1>
[redos@Redos ~]$ curl http://localhost:1380/123
<html>
<head><title>404 Not Found</title></head>
<body>
<center><h1>404 Not Found</h1></center>
<hr><center>nginx/1.28.2</center>
</body>
</html>
[redos@Redos ~]$ cat /var/log/myapp/access.log
127.0.0.1 - - [24/Apr/2026:12:08:31 +0300] "GET / HTTP/1.1" 200 21 "-" "curl/8.0.1" "-"
127.0.0.1 - - [24/Apr/2026:12:08:41 +0300] "GET /123 HTTP/1.1" 404 153 "-" "curl/8.0.1" "-"
[redos@Redos ~]$ cat /var/log/myapp/error.log
2026/04/24 12:05:43 [notice] 32546#32546: using the "epoll" event method
2026/04/24 12:05:43 [notice] 32546#32546: nginx/1.28.2
2026/04/24 12:05:43 [notice] 32546#32546: OS: Linux 6.12.69-1.red80.aarch64
2026/04/24 12:05:43 [notice] 32546#32546: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/04/24 12:05:43 [notice] 32547#32547: start worker processes
2026/04/24 12:05:43 [notice] 32547#32547: start worker process 32548
2026/04/24 12:05:43 [notice] 32547#32547: start worker process 32549
2026/04/24 12:08:41 [error] 32549#32549: *2 open() "/opt/myapp/public/123" failed (2: No such file or directory), client: 127.0.0.1, server: _, request: "GET /123 HTTP/1.1", host: "localhost:1380"
[redos@Redos ~]$
```

Проверили, что логи пишутся

```
[redos@Redos backend]$ sudo cat main.py
from fastapi import FastAPI
from datetime import datetime

app = FastAPI()

@app.get("/get_date")
async def get_date():
    return {"timestamp": datetime.now().isoformat()}
[redos@Redos backend]$
```

Создали небольшой код, который возвращает время


```
[redos@Redos backend]$ curl http://localhost:9000/get_date  
{"timestamp":"2026-04-24T12:15:26.258155"}[redos@Redos backend]$
```

Проверили, что все работает

```
server {  
    listen      1380;  
    listen      [::]:1380;  
    server_name _;  
    root        /opt/myapp/public;  
  
    location /api/ {  
        proxy_pass http://localhost:9000/;  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
    }  
    # Load configuration files for the default server block.  
    include /etc/nginx/default.d/*.conf;  
}
```

добавили новый блок location для проксирования запросов, начинающихся с /api/

```
[redos@Redos ~]$ curl http://localhost:1380/api/get_date  
<html>  
<head><title>502 Bad Gateway</title></head>  
<body>  
<center><h1>502 Bad Gateway</h1></center>  
<hr><center>nginx/1.28.2</center>  
</body>  
</html>
```

nginx работает, но выдает ошибку 502

```
[redos@Redos ~]$ sudo tail /var/log/myapp/error.log  
sudo ausearch -m avc -ts recent | grep nginx  
2026/04/24 12:41:50 [notice] 68887#68887: nginx/1.28.2  
2026/04/24 12:41:50 [notice] 68887#68887: OS: Linux 6.12.69-1.red80.aarch64  
2026/04/24 12:41:50 [notice] 68887#68887: getrlimit(RLIMIT_NOFILE): 1024:524288  
2026/04/24 12:41:50 [notice] 68896#68896: start worker processes  
2026/04/24 12:41:50 [notice] 68896#68896: start worker process 68897  
2026/04/24 12:41:50 [notice] 68896#68896: start worker process 68898  
2026/04/24 12:42:04 [crit] 68898#68898: *1 connect() to [::1]:9000 failed (13: Permission denied) while connecting to upstream, client: 127.0.0.1, server: _, request:  
"GET /api/get_date HTTP/1.1", upstream: "http://[::1]:9000/get_date", host: "localhost:1380"  
2026/04/24 12:42:04 [warn] 68898#68898: *1 upstream server temporarily disabled while connecting to upstream, client: 127.0.0.1, server: _, request: "GET /api/get_dat  
e HTTP/1.1", upstream: "http://[::1]:9000/get_date", host: "localhost:1380"  
2026/04/24 12:42:04 [crit] 68898#68898: *1 connect() to 127.0.0.1:9000 failed (13: Permission denied) while connecting to upstream, client: 127.0.0.1, server: _, requ  
est: "GET /api/get_date HTTP/1.1", upstream: "http://127.0.0.1:9000/get_date", host: "localhost:1380"  
2026/04/24 12:42:04 [warn] 68898#68898: *1 upstream server temporarily disabled while connecting to upstream, client: 127.0.0.1, server: _, request: "GET /api/get_dat  
e HTTP/1.1", upstream: "http://127.0.0.1:9000/get_date", host: "localhost:1380"  
type=SYSCALL msg=audit(1777023724.881:1525): arch=c00000b7 syscall=203 success=no exit=-13 a0=b a1=aaaaacd27d8b0 a2=1c a3=1 items=0 ppid=68896 pid=68898 auid=429496729  
5 uid=976 gid=976 euid=976 suid=976 fsuid=976 egid=976 sgid=976 fsgid=976 tty=(none) ses=4294967295 comm="nginx" exe="/usr/sbin/nginx" subj=system_u:system_r:ht  
tpd_t:s0 key=(null)  
type=AVC msg=audit(1777023724.881:1525): avc: denied { name_connect } for pid=68898 comm="nginx" dest=9000 scontext=system_u:system_r:ht  
tpd_t:s0 tcontext=system_u:  
object_r:http_port_t:s0 tclass=tcp_socket permissive=0  
type=SYSCALL msg=audit(1777023724.901:1526): arch=c00000b7 syscall=203 success=no exit=-13 a0=b a1=aaaaacd27d910 a2=10 a3=1 items=0 ppid=68896 pid=68898 auid=429496729  
5 uid=976 gid=976 euid=976 suid=976 fsuid=976 egid=976 sgid=976 fsgid=976 tty=(none) ses=4294967295 comm="nginx" exe="/usr/sbin/nginx" subj=system_u:system_r:ht  
tpd_t:s0 key=(null)  
type=AVC msg=audit(1777023724.901:1526): avc: denied { name_connect } for pid=68898 comm="nginx" dest=9000 scontext=system_u:system_r:ht  
tpd_t:s0 tcontext=system_u:  
object_r:http_port_t:s0 tclass=tcp_socket permissive=0  
[redos@Redos ~]$
```

```
[redos@Redos ~]$ sudo setsebool -P httpd_can_network_connect 1
[redos@Redos ~]$ sudo systemctl restart nginx
[redos@Redos ~]$ curl http://localhost:1380/api/get_date
{"timestamp":"2026-04-24T12:44:02.406938"}[redos@Redos ~]$
```

инициировали сетевое подключение, все заработало